

Self-Built Simple Quadruped Robot Dog - Midterm Progress Report

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

Applicant	CHENG-HSUAN LEE
Project	Self-Built Simple Quadruped Robot Dog
Course	Microprocessor Controlled Systems
Document Type	Midterm progress report
Primary Role	Team lead, mechanical designer, and controls developer
Original Source	robot-dog-midterm-report.pdf (20 pages)

Executive Summary

The midterm stage documented early kinematics work, hardware planning, circuit layout, and next-step development for the robot dog.

The value of this report is process evidence: the team moved from concept and simulation toward a physical platform that could later support real demonstrations.

Applicant Contribution Signals

- Used early MATLAB kinematics to guide real servo and linkage decisions.
- Planned electronics around ESP32, PCA9685, servos, and power delivery.
- Identified integration issues before the final demonstration stage.

Midterm Engineering State

The project was still transitioning from planned geometry to reliable physical motion.

Key work included gait reasoning, structure iteration, circuit planning, and preparing for calibration.

Next Development Goals

Complete body assembly, validate servo directions and offsets, improve power layout, and convert gait logic into embedded behavior.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Early gait concept	Used MATLAB reasoning	Prepared IK and reachable workspace checks	Kinematics
Electronics planning	Mapped controller, driver, and servo layout	Reduced final wiring confusion	Embedded design
Final demo preparation	Identified calibration and assembly tasks	Made remaining work actionable	Project leadership

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.